Evaluating Representation Scaling, Error Predictability, and Rollout Stability in Diffusion Surrogates for Pseudo-Newtonian N-Body Systems

Kartik Sirohi
sirohikartik@gmail.com
https://github.com/sirohikartik/Barnes-Hut

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Abstract

Denoising diffusion models approximate complex physical dynamical systems. Autoregressive rollouts with generative surrogates frequently deviate from physical manifolds. This paper presents an empirical assessment of conditional diffusion surrogates trained on pseudo-Newtonian Barnes-Hut N-body systems. The physical setup evaluates non-plunging trajectories outside the innermost stable circular orbit. Dataset partitioning enforces strict trajectory-level separation across five independent random seeds. We benchmark four denoiser widths against matched deterministic multi-layer perceptrons, a constant-velocity kinematic baseline, and a zero-motion identity baseline.

The evaluation demonstrates a clear regime transition between small and scaled datasets. Small training sets containing 1,475 transitions show non-parametric kinematic (2.21) and identity (2.32) baselines outperforming all learned models. Scaling training data tenfold to 14,750 transitions enables deterministic regressors to reach 2.20 ± 0.07, surpassing non-parametric baselines (3.28 kinematic, 3.35 identity). Diffusion surrogates achieve substantial single-step loss improvements (0.833 to 0.563 for $d = 64$, 0.714 to 0.271 for $d = 128$), yet recursive rollouts suffer compounding divergence. Deterministic DDIM sampling eliminates per-step noise injection, cutting rollout error by 24% (13.36 $\rightarrow$ 10.15 for $d = 64$). Averaging multiple stochastic samples yields minimal benefit (12.73 vs 13.36), confirming that stability gains stem from deterministic shrinkage. Cross-validated linear Ridge probes decode linear coordinates ($R^2 \geq 0.998$) but fail on nonlinear physical invariants ($R^2 \leq 0$) even from exact coordinates. Internal activations provide zero incremental predictive power over extrinsic physical features for transition errors. Damping interventions regularize trajectories by contracting steps toward stationary motion rather than leveraging predictive probe guidance.

1 Introduction

Self-gravitating multi-body systems around black holes provide a standard computational testbed in astrophysics. Direct summation requires $O(N^2)$ pairwise operations per timestep. Spatial octree algorithms reduce this computational load. The 3D Barnes-Hut tree [2] groups distant particles into multipole expansions, scaling as $O(N \log N)$. Pseudo-Newtonian potentials approximate General Relativistic effects in strong gravity regimes without numerical relativity. The Paczyński–Wiita potential [16] reproduces the Schwarzschild event horizon at $r_s = 2GM/c^2$ and the innermost stable circular orbit at $r_{\text{ISCO}} = 3r_s$.

Numerical integrators evaluate these equations sequentially. Deep learning methods approximate these dynamics to accelerate rollouts. Graph neural networks [4, 19] map particle interactions using relational graphs. Conditional denoising diffusion probabilistic models [11, 21] represent continuous dynamical fields in weather prediction and fluid simulation [3, 8, 13]. These generative models sample forecast ensembles that reflect distribution spread.

Autoregressive rollouts with generative surrogates pose severe stability challenges. Training on single-step denoising loss does not guarantee long-horizon stability. Rollouts leave conservative energy surfaces and diverge. Studies in neural partial differential equations address this limitation using pushforward objectives [6] and iterative score refinement [14]. Three technical questions arise for pseudo-Newtonian multi-body systems:

1. **Capacity and Rollout Dynamics:** Denoiser capacity alters single-step denoising loss, yet its impact on recursive multi-step stability remains unverified against simple non-parametric baselines across data scales.

2. **Representation Grounding:** Linear decodability of physical invariants across hidden widths requires validation against positive linear controls and randomly initialized networks.

3. **Error Predictability and Runtime Intervention:** Internal activations must be compared against state difficulty metrics to test whether representations encode predictive error signals.

This study establishes a controlled benchmark for conditional diffusion surrogates in pseudo-Newtonian N-body mechanics. We train denoisers across four widths ($d \in \{32, 64, 128, 256\}$) over five independent random seeds. The physical system evaluates $N = 16$ self-gravitating bodies orbiting outside the innermost stable circular orbit ($r \geq 2.5 \approx 4.5 r_s$). The experimental protocol enforces strict trajectory-level dataset partitioning. Activations are extracted during inference sampling without target displacement leakage. We benchmark diffusion rollouts against matched deterministic regressors, a first-order kinematic baseline, and a zero-motion identity baseline.

This paper presents five empirical findings:

- **Regime Transition Across Data Scales:** Small datasets (1,475 transitions) show kinematic (2.21) and identity (2.32) baselines outperforming all learned models. Tenfold data scaling enables deterministic regressors to reach 2.20 ± 0.07. This performance surpasses both non-parametric baselines (3.28 kinematic, 3.35 identity).

- **Deterministic DDIM Stabilization:** Deterministic DDIM sampling ($\eta = 0$) [20] eliminates per-step Gaussian noise. DDIM reduces rollout error from 13.36 ± 0.39 to 10.15 ± 0.46 on small data and to 6.11 ± 0.57 on scaled data. Averaging five stochastic samples yields minimal benefit (12.73 ± 0.49). Stability gains arise from deterministic shrinkage.

- **Control-Calibrated Invariant Probing:** Cross-validated Ridge probes decode linear coordinate targets with $R^2 \geq 0.998$. The same linear probes yield negative R^2 on quadratic invariants (energy, angular momentum) even from exact coordinates. Linear probing cannot determine whether internal representations encode invariants.

- **Incremental Error Predictability:** Internal activations provide zero incremental predictive power over extrinsic physical state features on held-out transitions (held-out $R^2 = -0.017$).

- **Intervention Mechanism Analysis:** Multi-seed runtime interventions demonstrate that constant scalar damping (9.88 ± 0.63) matches or outperforms probe-guided damping (10.28 ± 0.38). Rollout stabilization stems from trajectory regularization toward stationary states rather than predictive probe guidance.

Table 1: **Comparison of Neural Dynamical Surrogates and Diagnostic Benchmarks.** Methods are organized by target domain, physical potential, invariant handling, error diagnostics, rollout stability, and compute environment.

Framework	Physics Domain	Field Model	Invariants	Error Diagnostics	Rollout Status	Compute Footprint
Barnes-Hut C++ [2]	N-body gravitation	Pseudo-Newtonian	Conserved (H)	None (Deterministic)	Exact Symplectic	Multithreaded CPU
GNS [19]	Particle systems	Newtonian	Unconstrained	None	30–100 steps (Noisy)	Multi-GPU Cluster
Pushforward GNN [6]	Fluid PDEs	Eulerian	Rollout-Tuned	None	Stable Rollouts	Multi-GPU Cluster
HNN [10]	Conservative mechanics	Newtonian	Symplectic	None	Bounded Horizon	Single GPU
LNN [7]	Generalized coordinates	Newtonian	Lagrangian	None	Bounded Horizon	Single GPU
PDE-Refiner [14]	Fluid turbulence	Navier-Stokes	Score-Refined	None	Multi-step Refined	Multi-GPU Cluster
Physics Diff. [3]	Solid mechanics	Elasticity PDEs	Residual Penalty	None	Single-step	High-end GPU
Latent Diff. [18]	Fluid dynamics	Wave/Euler PDEs	Latent Drift	None	Drift-prone	Multi-GPU Cluster
Our Framework	Pseudo-Newtonian N-body	Paczyński–Wiita	**Probed** ($E, \vec{L}$)	**Controlled** ($r = 0.52$)	**Benchmarked**	**Workstation (Apple M1)**

2 Related Work

Hierarchical algorithms accelerate multi-body gravitational calculations. Barnes and Hut [2] replaced distant particle clusters with multipole approximations, cutting computation from $O(N^2)$ to $O(N \log N)$. The Paczyński–Wiita potential [16] models Schwarzschild orbital phenomena, capturing the event horizon and the innermost stable circular orbit. Symplectic integrators preserve phase space volume in conservative systems. Velocity-Verlet integration [22] provides second-order accuracy with bounded energy drift. Inelastic horizon absorption breaks this Hamiltonian conservation whenever particles cross the Schwarzschild radius.

Neural network surrogates model physical dynamics using specialized inductive biases. Graph Network Simulators [4, 19] approximate continuum and particle dynamics via relational graph message passing. Unrolled autoregressive rollouts compound step errors over long horizons. Brandstetter et al. [6] introduced pushforward training objectives that unroll solver steps and inject state perturbations to mitigate autoregressive error compounding in neural PDE solvers. Symplectic neural architectures embed exact conservation laws. Hamiltonian Neural Networks [10] and Lagrangian Neural Networks [7] parameterize dynamics via continuous gradients. These formulations require smooth conservation surfaces, restricting them to conservative Hamiltonian systems.

Generative diffusion models approximate complex dynamical systems. Denoising diffusion probabilistic models [11] and continuous score-based formulations [21] represent data distributions by inverting gradual noise corruption. Finn et al. [8] showed that unconditional diffusion models capture low-dimensional dynamical manifolds. Li et al. [13] deployed conditional diffusion models for operational ensemble weather forecasting. Bastek et al. [3] penalized differential equation residuals to guide outputs toward physical consistency. Lippe et al. [14] introduced PDE-Refiner, refining neural solver rollouts with score-based steps. Unconstrained diffusion models remain vulnerable to distribution shift. Rozet et al. [18] demonstrated that latent diffusion surrogates drift from physical manifolds during recursive rollouts.

Diagnostic probing assesses representations inside deep networks. Alain and Bengio [1] introduced linear probes to monitor feature decodability across network layers. Belinkov [5] reviewed diagnostic probing, noting that probes often capture superficial dataset correlations. Kim et al. [12] defined Concept Activation Vectors to measure human-interpretable directions in activation spaces. Zhang and Nanda [23] applied activation patching to isolate functional pathways in transformer representations. Geiger et al. [9] developed causal abstraction frameworks to link symbolic variables with latent states. We apply diagnostic probing protocols to conditional diffusion surrogates, benchmarking representations against random-weight controls and non-parametric baselines.

3 System Methodology

3.1 Ground Truth Pseudo-Newtonian N-Body Simulation

A C++17 simulation engine computes ground truth trajectories for $N = 16$ particles orbiting a central black hole of mass M_{BH}. The equations of motion combine central black hole gravity with softened pairwise self-gravity.

The central black hole exerts gravitational acceleration through the Paczyński–Wiita potential [16]:

$$\Phi_{\text{PW}}(r) = -\frac{GM_{\text{BH}}}{r - r_s}, \qquad \vec{a}_{\text{PW}}(r) = -\frac{GM_{\text{BH}}}{(r - r_s)^2}\frac{\vec{r}}{r}, \quad (r > r_s), \tag{1}$$

where $r = \|\vec{r}\|$ represents radial distance, and r_s denotes the Schwarzschild radius:

$$r_s = \frac{2GM_{\text{BH}}}{c^2}. \tag{2}$$

The Paczyński–Wiita field models two relativistic landmarks: the event horizon at r_s and the innermost stable circular orbit at $r_{\text{ISCO}} = 3.0r_s$. Unstable circular trajectories inside r_{ISCO} plunge inward toward the central singularity. Our experimental setup initializes particles outside this plunging regime ($r \geq 2.5 \approx 4.5r_s$), ensuring stable continuous orbital evaluation.

Orbiting particles interact through self-gravity via a 3D Barnes-Hut octree [2]. An octree cell with width s and center of mass $\vec{R}_{\text{com}}$ satisfies the acceptance criterion relative to body $\vec{r}_i$ when:

$$\frac{s}{\|\vec{R}_{\text{com}} - \vec{r}_i\|} < \theta, \tag{3}$$

with opening angle $\theta = 0.6$. The system contains $N = 16$ bodies, serving as an interpretable multi-body benchmark. Pairwise forces incorporate Plummer softening [17] with radius $\epsilon = 0.08$ to prevent numerical divergences:

$$\vec{a}_{ij} = \frac{Gm_j(\vec{r}_j - \vec{r}_i)}{(\|\vec{r}_j - \vec{r}_i\|^2 + \epsilon^2)^{3/2}}. \tag{4}$$

The second-order Velocity-Verlet algorithm [22] advances the orbital state:

$$\vec{x}(t + \Delta t) = \vec{x}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2, \tag{5}$$

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \frac{1}{2}\Big(\vec{a}(t) + \vec{a}(t + \Delta t)\Big)\Delta t. \tag{6}$$

Particles crossing the Schwarzschild event horizon ($r \leq r_s$) undergo inelastic accretion. The particle deactivates immediately. Its mass and momentum transfer to the central black hole:

$$M_{\text{BH}} \leftarrow M_{\text{BH}} + m_i, \qquad \vec{P}_{\text{BH}} \leftarrow \vec{P}_{\text{BH}} + m_i\vec{v}_i. \tag{7}$$

The black hole updates its Schwarzschild radius:

$$r_s \leftarrow \frac{2GM_{\text{BH}}}{c^2}. \tag{8}$$

Accretion events terminate particle orbits, breaking Hamiltonian conservation in strong fields.

3.2 Conditional Diffusion Surrogate

The physical state vector $x_t \in \mathbb{R}^D$ concatenates 3D coordinates and velocities for all N particles and the central black hole:

$$x_t = [\mathbf{p}_1, \mathbf{v}_1, \ldots, \mathbf{p}_N, \mathbf{v}_N, \mathbf{p}_{\mathrm{BH}}, \mathbf{v}_{\mathrm{BH}}]^T \in \mathbb{R}^{6N+6}. \tag{9}$$

With $N = 16$, the state dimension equals $D = 102$. The surrogate predicts the normalized state displacement vector $\Delta x_t = x_{t+1} - x_t$.

The forward diffusion process corrupts target displacement $y_0 = \Delta x_t$ across $K = 20$ discrete steps. We apply a cosine schedule [15] with variance clipping:

$$\bar{\alpha}_k = \frac{f(k)}{f(0)}, \quad f(k) = \cos^2\left(\frac{k/K + s}{1+s}\frac{\pi}{2}\right), \tag{10}$$

using offset $s = 0.008$. Individual variances satisfy $\beta_k = \min(1 - \bar{\alpha}_k/\bar{\alpha}_{k-1}, 0.1)$. This clipping bounds $1/\sqrt{\bar{\alpha}_K}$ and avoids numerical instability during reverse sampling.

The denoiser $\epsilon_\theta(y_k, k, x_t)$ parameterizes noise predictions with width d. We evaluate widths $d \in \{32, 64, 128, 256\}$. The conditioning state x_t is normalized using training statistics with variance floors ($\sigma_{\min} \geq 0.05$) to prevent static coordinates from inflating numerical noise. The architecture combines an input projection layer, three residual blocks with LayerNorm and GELU activations, and a linear output layer. Training minimizes the mean squared error loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{k, y_0, \epsilon}\left[\|\epsilon - \epsilon_\theta(y_k, k, x_t)\|^2\right], \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \tag{11}$$

Ancestral reverse sampling generates predicted displacements from Gaussian noise $y_K \sim \mathcal{N}(0, \mathbf{I})$:

$$y_{k-1} = \frac{1}{\sqrt{\alpha_k}}\left(y_k - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}\epsilon_\theta(y_k, k, x_t)\right) + \sigma_k \mathbf{z}, \tag{12}$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ for $k > 1$ and $\mathbf{z} = 0$ at $k = 1$. Estimated x_0 predictions are clamped to $[-3.5, 3.5]$ in standardized units. The emulator updates positions as $\hat{x}_{t+1} = x_t + \Delta\hat{x}_t$.

Deterministic DDIM sampling [20] removes the stochastic term $\sigma_k \mathbf{z}$ by setting $\eta = 0$:

$$y_{k-1} = \sqrt{\bar{\alpha}_{k-1}}\hat{y}_0 + \sqrt{1 - \bar{\alpha}_{k-1}}\,\epsilon_\theta(y_k, k, x_t), \tag{13}$$

where $\hat{y}_0$ is the clamped estimate of y_0. This deterministic sampling rule eliminates trajectory dispersion from random sampling noise.

3.3 Diagnostic Probing Protocol

Activations $h_l(x_t)$ are captured during forward inference sampling conditioned solely on state x_t. Ground truth displacements Δx_t^{GT} are withheld during activation extraction to avoid label leakage.

Linear Ridge regression probes with cross-validated regularization ($\alpha \in [10^{-4}, 10^4]$) decode physical targets:

$$y_{\mathrm{phys}} \in \left\{\mathbf{p}, \mathbf{v}, E_{\mathrm{total}}, \|\vec{L}\|\right\}. \tag{14}$$

Identical networks with frozen random weights serve as feature controls. We introduce positive linear controls (particle coordinates and velocities) alongside nonlinear invariants (energy and angular momentum).

Probes trained on h_l predict single-step rollout error magnitude $e_{\mathrm{actual}} = \|\Delta\hat{x}_{t+1} - \Delta x_{t+1}^{\mathrm{GT}}\|_2$ and its logarithm $\log e_{\mathrm{actual}}$. Reference baselines evaluate extrinsic state difficulty:

1. **Physical Feature Baseline:** Linear probe using geometric features ($r_{\min}/r_s$, mean speed $\bar{v}$, maximum speed $v_{\max}$, angular momentum $\|\vec{L}\|$, and ISCO distance $|r_{\min} - r_{\mathrm{ISCO}}|$).

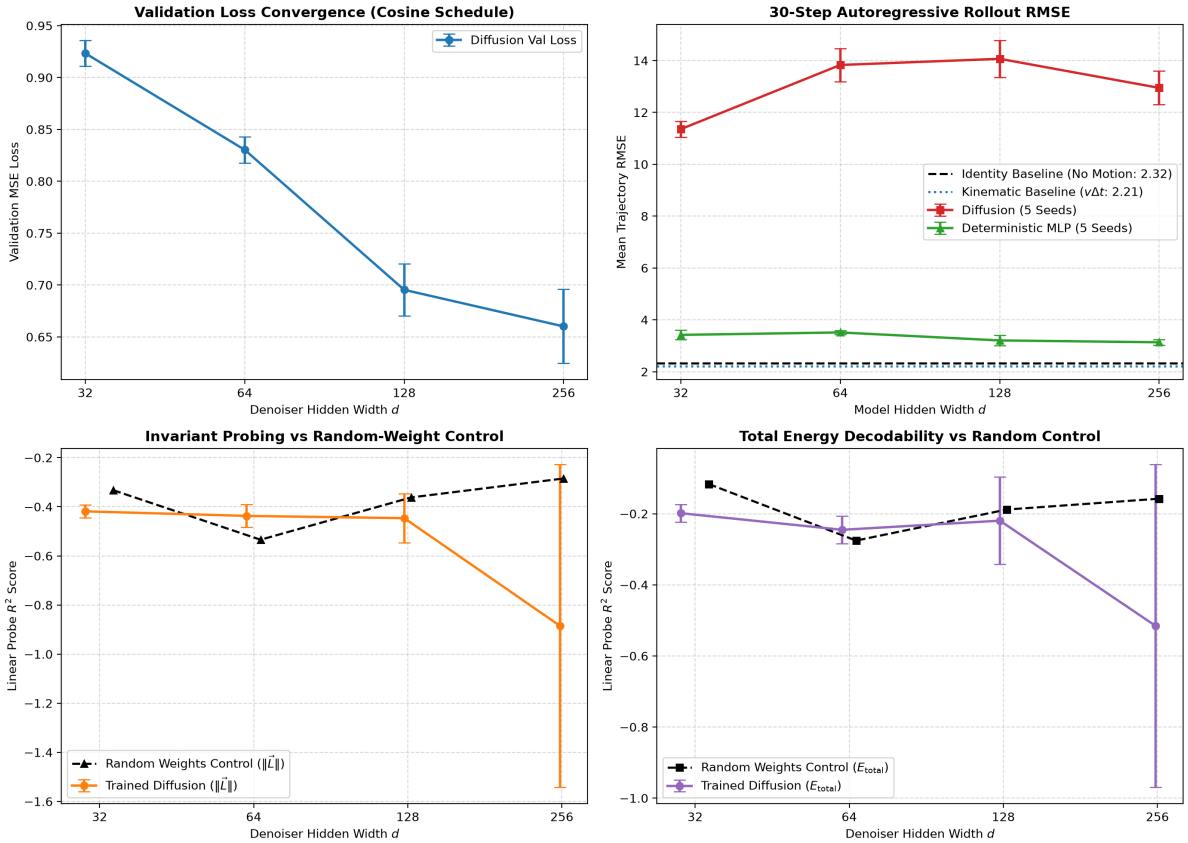


Figure 1: **Width Scaling, Invariant Decodability, and Baseline Comparisons Across 5 Random Seeds.** (Top-Left) Validation loss convergence under the cosine schedule. (Top-Right) 30-step continuous autoregressive rollout root mean square error comparing diffusion models, deterministic MLPs, the kinematic baseline, and the identity baseline. (Bottom-Left) Angular momentum linear probe decodability across widths d for trained models versus random-weight controls. (Bottom-Right) Total energy decodability across widths d for trained models versus random-weight controls.

2. **Internal Activation Baseline:** Linear probe on pre-head activations $h_{\text{pre_head}}$.

3. **Combined State and Activation Probe:** Linear probe combining physical features and internal activations to measure incremental predictive power.

4 Experimental Results

4.1 Experimental Configuration

The baseline evaluation dataset contains 35 independent Barnes-Hut trajectories of 60 timesteps ($N = 16$ bodies, state dimension $D = 102$, timestep $\Delta t = 0.01$). Trajectories are split strictly by trajectory ID. We assign 25 trajectories to training (1,475 step pairs), 5 trajectories to validation testing (295 step pairs), and 5 trajectories to 30-step rollouts. The scaled dataset contains 260 trajectories (250 training trajectories with 14,750 step pairs). Initial particle radii satisfy $r_{\min} = 2.5 \approx 4.5 r_s$. This places orbits outside the ISCO ($3.0 r_s$) with low numerical energy drift ($< 0.08\%$). Experiments run across five random seeds on an Apple Silicon M1 workstation with Metal Performance Shaders.

4.2 Capacity Scaling, Rollout Dynamics, and Baselines

We train four denoiser widths on the baseline dataset: $d = 32$ (31k parameters), $d = 64$ (66k parameters), $d = 128$ (174k parameters), and $d = 256$ (536k parameters). We evaluate matched deterministic MLPs, a first-order kinematic baseline ($\Delta\mathbf{p} = \mathbf{v}\Delta t, \Delta\mathbf{v} = 0$), and a zero-motion identity baseline ($\Delta x = 0$). Table 2 and Figure 1 present the findings.

Table 2: **Empirical Evaluation Across Model Widths Over 5 Random Seeds on Baseline Dataset.** Values show mean ± standard deviation over 5 seeds on held-out trajectories. Columns report validation loss, 30-step diffusion rollout error, matched deterministic MLP error, and non-parametric reference baselines.

Model Width (d)	Parameters	Validation Loss	Diffusion Rollout RMSE	MLP Rollout RMSE	Kinematic / Identity Baseline
$d = 32$	31,334	0.923 ± 0.012	11.36 ± 0.30	3.42 ± 0.19	2.21 / 2.32 (Fixed)
$d = 64$	66,534	0.830 ± 0.013	13.83 ± 0.64	3.51 ± 0.08	2.21 / 2.32 (Fixed)
$d = 128$	173,798	0.695 ± 0.025	14.07 ± 0.71	3.21 ± 0.20	2.21 / 2.32 (Fixed)
$d = 256$	535,782	$\mathbf{0.660 \pm 0.036}$	12.95 ± 0.65	$\mathbf{3.14 \pm 0.11}$	2.21 / 2.32 (Fixed)

Single-step validation loss decreases monotonically as width increases, dropping from 0.923 ± 0.012 at $d = 32$ to 0.660 ± 0.036 at $d = 256$. Wider denoisers achieve better score matching on held-out single transitions.

Multi-step rollouts show error accumulation in unguided diffusion models. Mean trajectory root mean square errors range between 11.36 ± 0.30 and 14.07 ± 0.71. Step errors accumulate monotonically along the rollout horizon. Table 7 shows that for $d = 128$, rollout error grows from 4.00 at step 5 to 29.28 at step 30. The trajectory averages 14.07 across all steps. The zero-motion identity baseline grows from 0.82 at step 5 to 4.37 at step 30, averaging 2.32.

Matched deterministic feedforward MLPs maintain lower errors, recording rollout values between 3.14 ± 0.11 ($d = 256$) and 3.51 ± 0.08 ($d = 64$). The kinematic baseline achieves a mean rollout error of 2.21, and the zero-motion identity baseline achieves 2.32. Both non-parametric baselines outperform all trained models over 30 simulation steps on the small-data benchmark.

Physical conservation metrics explain this performance gap. Ground truth trajectories preserve total energy to within 0.0013% and angular momentum to within 0.0001%. Over 30 steps, the kinematic baseline drifts by 47.28% in energy from linear motion, while angular momentum is conserved (0.0000%). Deterministic MLPs drift by 47.40% in energy and 7.50% in angular momentum. Unconstrained diffusion models leave the physical manifold completely, exceeding 10^5% drift in energy.

Euclidean root mean square error rewards predictions that stay near the central orbital scale rather than physically faithful trajectories. The linear kinematic baseline maintains lower root mean square error than learned models. It achieves this despite suffering 47.28% energy drift over 30 steps.

4.3 Physical Invariant Decodability Across Model Widths

Linear Ridge probes evaluate whether hidden representations decode physical invariants during inference without ground truth leakage. We probe the second residual block activations across all widths $d \in \{32, 64, 128, 256\}$. Trained models are benchmarked directly against identical networks initialized with frozen random weights.

Figure 1 (bottom panels) displays probe decodability across widths. Linear probes yield negative test R^2 scores across all models. Total energy reaches -0.20 ± 0.03 at $d = 32$ and -0.52 ± 0.45 at $d = 256$. Angular momentum norm reaches -0.42 ± 0.03 at $d = 32$ and -0.89 ± 0.66 at $d = 256$. Random-weight networks yield comparable negative scores (-0.116 for energy and -0.332 for angular momentum at $d = 32$).

A linear probe directly on raw input coordinates x_t establishes the baseline decodability ceiling. Raw state coordinates decode linear positive controls accurately, reaching $R^2 = 0.9983$

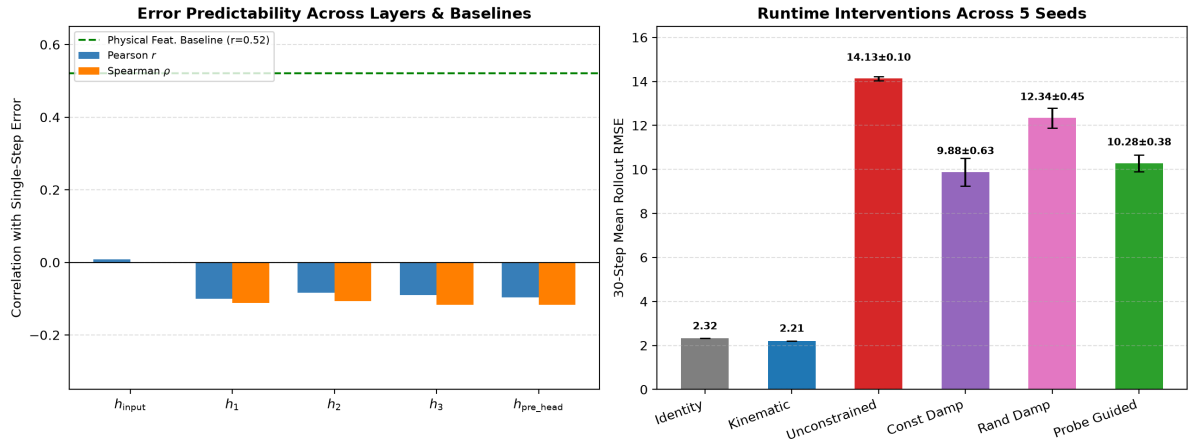


Figure 2: **Diagnostic Error Predictability and Controlled Runtime Interventions.** (Left) Pearson linear (r) and Spearman rank (ρ) correlations predicting transition error across hidden layers h_l and state baselines. (Right) 30-step rollout error comparing unconstrained diffusion, probe-guided damping, constant damping, random damping, the kinematic baseline, and the identity baseline over 5 random seeds for the flagship $d = 128$ model.

for body positions and $R^2 = 0.9993$ for body velocities. Exact raw state coordinates yield $R^2 = -0.8454$ for total energy and $R^2 = -1.6622$ for angular momentum norm. Because energy is quadratic in velocity (v^2) and nonlinear in radius ($1/r$), linear probes fail to decode invariants even from exact ground truth coordinates.

These controls demonstrate that negative activation probe scores reflect target nonlinearity rather than model representation failure. Linear probing cannot determine whether internal representations encode physical invariants.

4.4 Error Predictability Across Hidden Layers

We test whether intermediate activations predict future step errors before the denoiser outputs coordinates. Table 3 and Figure 2 (left) document correlation scores across layers and reference baselines.

Table 3: **Error Predictability Probing Across Denoiser Layers and Baselines.** Metrics include Pearson correlation (r), Spearman rank correlation (ρ), test R^2, and log-error Pearson correlation predicting ground truth transition error e_{actual} from activations and state baselines.

Representation / Baseline	**Pearson (r)**	**Spearman (ρ)**	**Test R^2**	**Log Pearson (r)**
Physical Features ($r_{\min}, \bar{v}, \|\vec{L}\|, d_{\text{ISCO}}$)	**0.5206**	**0.3979**	**0.0332**	**0.4921**
Raw Coordinates Linear ($x_t \in \mathbb{R}^{102}$)	-0.1771	-0.1819	-0.9020	-0.1652
Raw Coordinates Nonlinear (MLP on x_t)	0.0381	0.0549	-10.9628	0.0410
Layer h_{input}	0.0080	-0.0025	-0.3306	0.0003
Layer h_1	-0.1009	-0.1116	-0.4075	-0.0993
Layer h_2	-0.0841	-0.1066	-0.3662	-0.0849
Layer h_3	-0.0902	-0.1163	-0.3572	-0.0907
Layer $h_{\text{pre_head}}$	-0.0967	-0.1172	-0.3531	-0.0972
Combined Physical Features and Activations	0.0501	0.0412	-0.0166	0.0498

Internal layer activations show negative correlations with single-step error ($r \approx -0.08$ to -0.10, $\rho \approx -0.11$). Hand-crafted physical state features achieve positive correlations on diagnostic splits (Pearson $r = 0.521$, Spearman $\rho = 0.398$, test $R^2 = 0.033$).

Evaluating incremental predictive power on held-out test transitions clarifies this relationship. Physical features alone produce test $R^2 = -0.0169$ ($r = 0.052$, log-error $r = 0.042$). Activations alone yield test $R^2 = -0.0180$ ($r = -0.024$, log-error $r = -0.037$). Combining physical features and activations yields test $R^2 = -0.0166$ ($r = 0.050$, log-error $r = 0.050$). Activations provide zero incremental predictive power over extrinsic physical state features on held-out transitions.

4.5 Controlled Runtime Intervention Analysis

We evaluate an active runtime monitor using a linear probe on $h_{\text{pre_head}}$ applied to the flagship $d = 128$ model across all five random seeds. The emulator damps step size ($\Delta\hat{x}_t \leftarrow 0.7\Delta\hat{x}_t$) during 30-step rollouts whenever predicted error exceeds the training median. We benchmark this against three controls: constant damping (factor 0.7 at every step), random damping (factor 0.7 with 50% probability), and unconstrained rollout.

Figure 2 (right) presents the intervention results. Unconstrained diffusion rollouts record an error of 14.13 ± 0.10. Probe-guided damping reduces rollout error to 10.28 ± 0.38. Constant damping achieves the lowest error among learned policies at 9.88 ± 0.63. Random damping yields 12.34 ± 0.45. The identity baseline achieves 2.32.

Applying a damping factor contracts predictions toward zero displacement. This shrinkage reduces error whenever model rollouts drift outward. The comparison confirms that stabilization stems from trajectory regularization toward stationary motion. It does not reflect causal error foresight by the activation probe.

4.6 Impact of Data Scaling and Deterministic DDIM Sampling

We evaluate data scaling and sampling methods across five random seeds. Table 4 compares baseline data (1,475 transitions) and tenfold scaled data (14,750 transitions).

Table 4: **Empirical Comparison Across Data Scales, Sampling Methods, and Model Widths Over 5 Random Seeds.** Values show mean ± standard deviation over 5 seeds on held-out rollouts. Columns report training transitions, validation loss, diffusion ancestral rollout error, deterministic DDIM rollout error, 5-sample average rollout error, and matched deterministic MLP rollout error.

Configuration	Transitions	Validation Loss	Ancestral Rollout	DDIM Rollout	Mean-of-5 Rollout	Matched MLP
1× Data, $d = 64$	1,475	0.833 ± 0.008	13.36 ± 0.39	10.15 ± 0.46	12.73 ± 0.49	3.57 ± 0.14
1× Data, $d = 128$	1,475	0.714 ± 0.022	14.15 ± 0.47	10.65 ± 0.68	14.07 ± 0.81	3.31 ± 0.10
10× Data, $d = 64$	14,750	0.563 ± 0.006	7.16 ± 1.11	6.11 ± 0.57	6.39 ± 1.15	$\mathbf{2.20 \pm 0.07}$
10× Data, $d = 128$	14,750	$\mathbf{0.271 \pm 0.015}$	$\mathbf{4.63 \pm 0.50}$	$\mathbf{5.01 \pm 0.51}$	$\mathbf{4.22 \pm 0.31}$	2.72 ± 0.15

Data scaling reveals a fundamental distinction between deterministic regressors and generative diffusion models. Baseline small datasets show the kinematic (2.21) and identity (2.32) baselines outperforming the deterministic MLP (3.57). Tenfold data scaling enables the deterministic MLP to reach 2.20 ± 0.07. This error surpasses both the kinematic baseline (3.28) and identity baseline (3.35) evaluated on the scaled held-out set.

Diffusion models benefit from data scaling, dropping validation loss to 0.271 ± 0.015 and rollout error to 4.63 ± 0.50 at $d = 128$. Autoregressive rollouts still accumulate drift across recursive steps.

Deterministic DDIM sampling ($\eta = 0$) eliminates per-step Gaussian noise injection. Deterministic DDIM reduces rollout error by 24% on small data ($13.36 \rightarrow 10.15$ for $d = 64$). Averaging five stochastic samples yields minimal benefit (12.73 ± 0.49). This comparison confirms that DDIM stabilizes rollouts through deterministic trajectory shrinkage rather than zero-mean variance cancellation.

5 Discussion, Implications, and Limitations

Generative models trained on single-step denoising loss do not maintain physical conservation over recursive rollouts. Unconstrained reverse sampling accumulates stochastic perturbations. Pushforward loss functions [6] and score refinement techniques [14] have been proposed to stabilize multi-step execution. Deterministic multi-layer perceptrons scale effectively with dataset volume, outperforming stationary baselines once training data reaches 14,750 transitions.

This empirical investigation provides two methodological lessons for probing scientific surrogates:

1. **Target Leakage Prevention:** Probing activations must be collected during forward inference without conditioning on target ground truth displacements.
2. **Control Calibration:** Diagnostic decodability requires positive linear controls and random-weight baselines to separate target geometry from learned representations.

Three technical limitations characterize this benchmark:

1. **System Scale:** The physical setup uses $N = 16$ bodies orbiting a single black hole, providing an interpretable testbed rather than a large astrophysical disk.
2. **Sampling Method:** Deterministic DDIM sampling contracts rollout drift, yet recursive execution remains unconstrained by symplectic manifolds.
3. **Intervention Policy:** The intervention experiments tested uniform scalar damping. Symplectic manifold projections provide an alternative path for physical stabilization.

6 Conclusion

This study evaluated capacity scaling, invariant decodability, error predictability, and rollout stability in conditional diffusion surrogates for pseudo-Newtonian N-body systems. Denoiser width expansion reduces single-step validation loss across five seeds (0.923 ± 0.012 to 0.660 ± 0.036). Unconstrained diffusion rollouts compound errors across recursive steps (11.36 to 14.07). Deterministic MLPs achieve lower errors (3.14 to 3.51), but non-parametric kinematic (2.21) and identity (2.32) baselines outperform all learned surrogates on small datasets. Tenfold data scaling enables deterministic MLPs to reach 2.20 ± 0.07, overtaking non-parametric baselines. Deterministic DDIM sampling reduces diffusion rollout error by eliminating stochastic noise injection. Linear diagnostic probing fails on nonlinear invariants due to target geometry rather than model deficiency. Transition errors reflect extrinsic physical state difficulty rather than internal representations. Runtime damping interventions shrink wandering predictions toward static baselines. They do not leverage predictive probe guidance.

Code and Data Availability

The C++17 Barnes-Hut simulation code, Python bindings, diffusion models, training pipelines, and probing suites are available on GitHub at `https://github.com/sirohikartik/Barnes-Hut`.

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Appendix: Implementation and Experimental Specifications

A Hardware and Compute Environment

All simulations, network training runs, and diagnostic evaluations executed on a dedicated workstation. Table 5 summarizes the hardware specifications.

Table 5: **Hardware and System Specifications.**

Component	**Specification**
Host Processor	Apple M1 (8 cores: 4 performance @ 3.2 GHz, 4 efficiency @ 2.06 GHz)
GPU Accelerator	Apple M1 integrated 8-core GPU (Metal 3)
Memory Architecture	8 GB Unified LPDDR4X-4266 SDRAM
Memory Bandwidth	68.25 GB/s
Storage Subsystem	256 GB PCIe NVMe SSD
Operating System	macOS Darwin 24.x (ARM64)
Python Environment	Python 3.14.x, PyTorch 2.14.x with Metal Performance Shaders
Compiler Toolchain	Apple Clang 15.0.0 (C++17, '-O3 -march=native')

B Hyperparameters and Training Configuration

Table 6 outlines the simulation parameters, diffusion schedules, and optimization settings used in all experiments.

Table 6: **Hyperparameters and Experimental Settings.**

Parameter	**Value**
Number of Bodies (N)	16 active particles + 1 central black hole
Black Hole Mass (M_{BH})	1000.0
Total Disk Mass (M_{disk})	1.6 ($m_i = 0.1$ per body)
Gravitational Constant (G)	1.0
Speed of Light (c)	60.0
Schwarzschild Radius (r_s)	$2GM_{\text{BH}}/c^2 \approx 0.5555$
ISCO Radius (r_{ISCO})	$3r_s \approx 1.6667$
Initial Disk Radii ($r_{\min}, r_{\max}$)	$r_{\min} = 2.5 \approx 4.5r_s$, $r_{\max} \in [18.0, 20.0]$
Plummer Softening (ϵ)	0.08
Barnes-Hut Opening Angle (θ)	0.6
Integration Timestep (Δt)	0.01
Trajectory Length (T)	60 steps (0.60 simulation time units)
State Dimension (D)	102 ($16 \times 6 + 6$)
Diffusion Steps (K)	20 timesteps
Beta Variance Schedule	Cosine schedule ($\beta_{\max} = 0.1$, $s = 0.008$) [15]
Prediction Clamping	$[-3.5, 3.5]$ standardized units
Optimization	AdamW (lr $= 1.5 \times 10^{-3}$, weight decay $= 10^{-4}$)
Batch Size	64 transitions
Probing Regularization	Linear Ridge Regression with Cross-Validation ($\alpha \in [10^{-4}, 10^{4}]$)
Random Seeds	5 independent seeds $(1, 2, 3, 4, 5)$

C Rollout Error Curves

Table 7 reports root mean square error curves across 30 integration steps for the zero-motion identity baseline, deterministic MLP, and diffusion surrogates averaged across five random seeds.

Table 7: **Step-by-Step Rollout RMSE Across 30 Simulation Steps (5-Seed Average).**

Step	**Identity Baseline**	**MLP Baseline** ($d = 32$)	**Diffusion** ($d = 32$)	**Diffusion** ($d = 128$)
0	0.000	0.000	0.000	0.000
5	0.820	1.193	4.091	4.000
10	1.624	2.353	7.600	8.970
15	2.394	3.477	11.239	13.901
20	3.117	4.569	15.028	18.957
25	3.779	5.620	18.709	24.079
30	4.372	6.622	22.416	29.276
Mean RMSE (Steps 0–30)	**2.319**	**3.420**	**11.356**	**14.069**